Maxwell's distribution

Probability distribution: $f(\bar{r},\bar{v})\, d^3\bar{r}\, d^3\bar{v}$ = number of molecules with positions between $\vec{r}$ and $\vec{r}+d\vec{r}$ and velocities in the range $\vec{v}$ and $\vec{v}+d\vec{v}$

$$\Rightarrow f(\bar{r},\bar{v})\, d^3\bar{r}\, d^3\bar{v} = C \underbrace{\exp\left(-\frac{mv^2}{2kT}\right)}_{\text{Boltzmann factor (ideal gas } E = K.E.)} d^3\bar{r}\, d^3\bar{v}$$

Normalization condition

$$C \int_{(V)}\int_{(V)} f(\bar{r},\bar{v})\, d^3\bar{r}\, d^3\bar{v} = N \quad (\#\text{ of molecules})$$

$$CV \underbrace{\int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2kT}} dv_x}\int e^{-\frac{mv_y^2}{2kT}} dv_y \int e^{-\frac{mv_z^2}{2kT}} dv_z = N$$

$$x = \sqrt{\frac{2kT}{m}}\, x : \int_{-\infty}^{\infty} e^{-x^2} dx \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2\pi kT}{m}}$$

But $$\int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{+\infty} e^{-y^2} dy = \int e^{-(x^2+y^2)} \underbrace{dx\,dy}_{2\pi r\, dr} =$$

$$= \pi \int_0^{\infty} e^{-r^2} dr^2 = \pi \Rightarrow \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\Rightarrow C = \frac{N}{V}\left(\frac{m}{2\pi kT}\right)^{3/2}$$

$$f(\bar{r},\bar{v})\, d^3\bar{r}\, d^3\bar{v} = \frac{N}{V}\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{mv^2}{2kT}}\, d^3\bar{r}\, d^3\bar{v}$$

Distribution is clearly uniform in space.

$\Rightarrow$ number of molecules in volume V with velocities $(\vec{v}, \vec{v}+d\vec{v})$

$$D(\bar{v})\, d^3\bar{v} = V f(\bar{v})\, d^3\bar{v} = N\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{mv^2}{2kT}}\, d^3\bar{v}$$

Distribution of a given component of velocity, say v_y:

$$g(v_y)\,dv_y = \int_{v_x}\int_{v_z} D(\vec{v})\,d^3\vec{v} =$$

$$= N\left(\frac{m}{2\pi kT}\right)^{3/2} \underbrace{\int e^{-\frac{m}{2kT}(v_x^2+v_z^2)}\,dv_x\,dv_z}_{\left[\left(\frac{2\pi kT}{m}\right)^{1/2}\right]^2} \cdot e^{-\frac{mv_y^2}{2kT}}\,dv_y$$

$$= N\left(\frac{m}{2\pi kT}\right)^{1/2} e^{-\frac{mv_y^2}{2kT}}\,dv_y$$

$$\Rightarrow \overline{v_y} = \frac{\int_{-\infty}^{\infty} v_y\, g(v_y)\,dv_y}{\underbrace{\int_{-\infty}^{\infty} g(v_y)\,dv_y}_{N}} = \left(\frac{m}{2\pi kT}\right)^{1/2}\int_{-\infty}^{\infty} v_y\, e^{-\frac{mv_y^2}{2kT}}\,dv_y = 0$$

$$\overline{v_y^2} = \left(\frac{m}{2\pi kT}\right)^{1/2}\int_{-\infty}^{\infty} v_y^2\, e^{-\frac{mv_y^2}{2kT}}\,dv_y = \left(\frac{m}{2\pi kT}\right)^{1/2}\left(\frac{2kT}{m}\right)^{3/2}\underbrace{\int_{-\infty}^{\infty} y^2 e^{-y^2}\,dy}_{\frac{\sqrt{\pi}}{2}}$$

$$v_y = \sqrt{\frac{2kT}{m}}\,y$$

$$= \frac{1}{\sqrt{\pi}}\,\frac{2kT}{m}\,\frac{\sqrt{\pi}}{2} = \frac{kT}{m}$$

$$\Rightarrow \frac{1}{2}m\overline{v_y^2} = \frac{1}{2}kT \Rightarrow \frac{1}{2}m\left(\overline{v_x^2}+\overline{v_y^2}+\overline{v_z^2}\right) = \frac{3}{2}kT$$

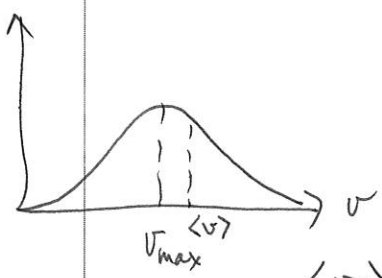
(equipartition Th)

Distribution of speeds = number of molecules with speed $v = |\vec{v}|$ in the range $(v, v+dv)$

$$F(v)\,dv = \int_{\text{(directions)}} f(\vec{v})\, d^3\vec{v} = \underbrace{4\pi v^2}_{\substack{\text{phase space} \\ \text{of molecules}}} N\, e^{-\frac{mv^2}{2kT}} dv \left(\frac{m}{2\pi kT}\right)^{3/2}$$

(Maxwell's speed distribution)

Mean speed $\langle v \rangle = \dfrac{\int_0^\infty v\, F(v)\,dv}{\underbrace{\int_0^\infty F(v)\,dv}_{N}}$



$$\langle v \rangle = 4\pi \int_0^\infty v \cdot v^2 e^{-\frac{mv^2}{2kT}} dv \left(\frac{m}{2\pi kT}\right)^{3/2} =$$

$$v = \sqrt{\frac{2kT}{m}}\, y$$

$$= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{2kT}{m}\right)^{\frac{4}{2}} \underbrace{\int_0^\infty y^3 e^{-y^2} dy}_{\frac{1}{2}} =$$

$$= 4\pi \frac{1}{\pi^{3/2}} \left(\frac{2kT}{m}\right)^{1/2} \frac{1}{2} = \sqrt{\frac{8}{\pi}\frac{kT}{m}}$$

Most probable speed? $\Rightarrow$ find maximum of distribution

$$0 = \frac{dF}{dv} = \frac{d}{dv}\left(v^2 e^{-\frac{mv^2}{2kT}}\right) = 2v e^{-\frac{mv^2}{2kT}} + v^2\left(-\frac{mv}{kT}\right) e^{-\frac{mv^2}{2kT}}$$

$$\Rightarrow 2v - \frac{mv^3}{kT} = 0 \Rightarrow v_{max}^2 = \frac{2kT}{m} \Rightarrow v_{max} = \sqrt{\frac{2kT}{m}}$$

Finally, $v_{rms} = \sqrt{\langle v^2 \rangle}$

$$\langle v^2 \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{2kT}{m}\right)^{5/2} \underbrace{\int_0^\infty y^4 e^{-y^2} dy}_{\frac{3\sqrt{\pi}}{8}} =$$

$$= 4\pi \left(\frac{1}{\pi^{3/2}}\right) \left(\frac{2kT}{m}\right) \left(\frac{3\sqrt{\pi}}{8}\right) = \frac{3kT}{m}$$

So that $\Rightarrow$ $\frac{1}{2} m\langle v^2 \rangle = \frac{3}{2} kT$ (as it should be from equipart.)

$$\Rightarrow v_{rms} = \sqrt{\frac{3kT}{m}}$$

~~Effusion~~